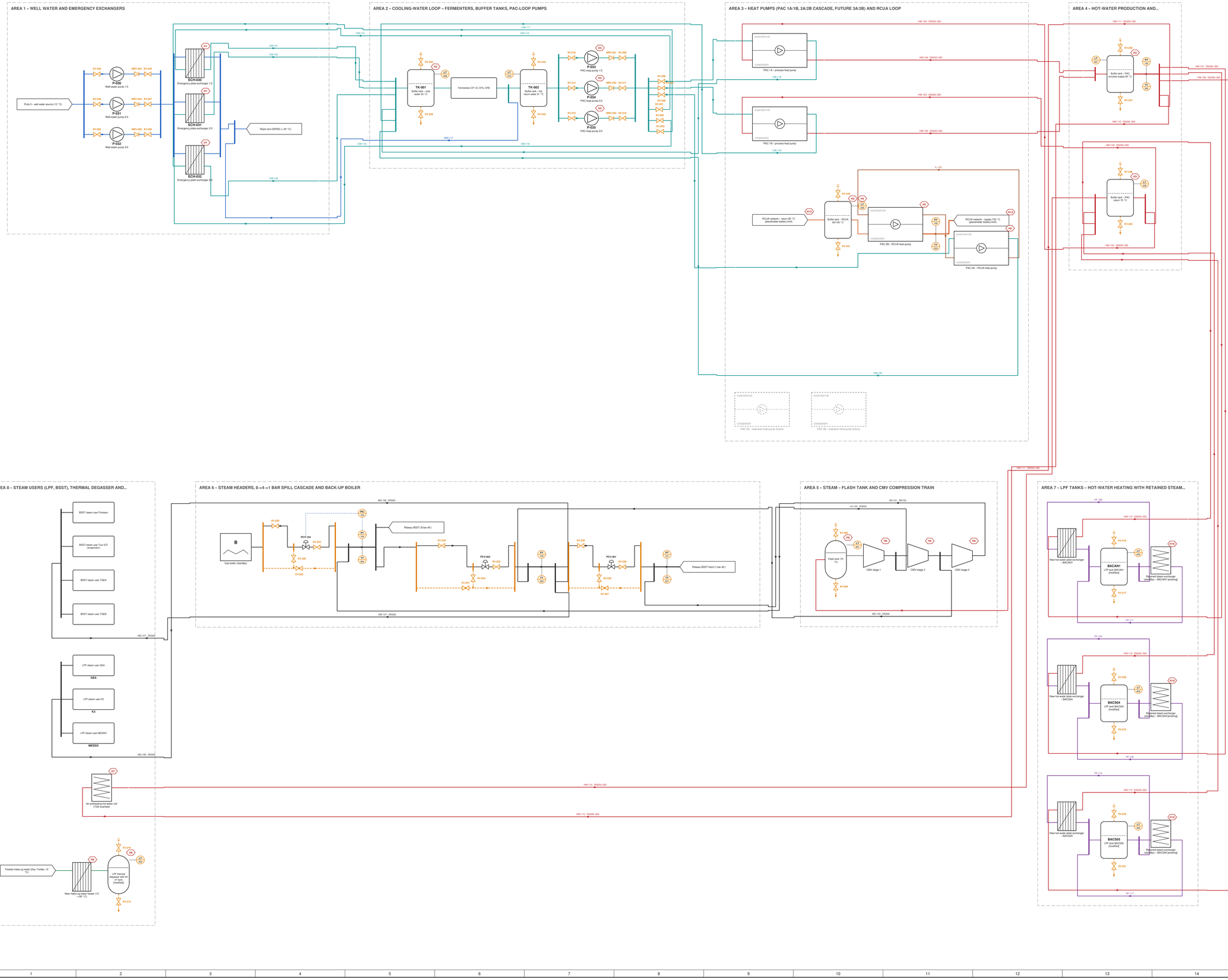


LEGEND

- Cooling water 20/31
- Flash steam
- Hot water process 90/70
- Process fluid (LPF tanks)
- Treated make-up water
- Well water
- Drain
- Hot water RCUA 85/105
- Internal loop PAC 2A/2B
- Steam
- Vent

- Isolation valve, technology to be selected (generic)
- Globe valve (by-pass)
- Instrument – ISA 5.1 function / loop number (e.g. PT 101)
- Proposed addition (rule-based, pending review)
- H1 – HOLD – decision needed, see hold list
- Check valve
- Control valve with actuator
- Centrifugal pump
- Future equipment (LifeCyclePhase = Future)
- Instrument signal

GENERAL NOTES

1. Draft generated from the specification 'Cahier des Charges P&ID Consolidé' – not for construction.
2. Orange items are additions proposed by the project rules (valves, instruments, tags, line numbers); approval is per rule, see the decision register.
3. Line numbers are «service code»-«sequence». Sizes and piping classes are not given in the specification and are not shown unless stated (DN).
4. Valve failure positions, design pressures, relief devices, instrument ranges and nozzle sizes are not given and are not assumed.
5. Future heat pumps PAC 3A/3B are shown dashed and unconnected (LifeCyclePhase = Future).
6. Operating modes (normal, emergency exchangers, ultimate) are realised with the isolation valves shown; see the operating-mode report.

HOLDS – DECISIONS REQUIRED BEFORE ISSUE

- **H1** **Emergency exchangers ECH-030..032 – parallel or series?** Drawn: Parallel branch of the cooling-water loop, beside the heat-pump evaporators. Question: Confirm parallel arrangement and the well-water path in Normal mode.
- **H2** **PAC 2A/2B – physical cascade?** Drawn: 2A condenser feeds the 2B evaporator through the 55/50 °C internal loop; only 2B delivers 85/105 °C to RCUA. Question: Confirm the 2A/2B cascade connections.
- **H3** **Circulation pumps of the fermenter leg and hot-water loops** Drawn: Only P-033..035 (TK-002 outlet). No other pump drawn – the specification names none. Question: Where are the circulation pumps of the fermenter leg and of the hot-water loops?
- **H4** **RCUA buffer – one tank or one per temperature level?** Drawn: One RCUA buffer tank (85/105 °C). Question: One RCUA tank or one per temperature level?
- **H5** **Spill valves 8→4 and 4→1 bar – control philosophy** Drawn: Control valves with manifolds, no control loop (variable, setpoint and fail position not given). Question: Control philosophy of the overproduction cascade.
- **H6** **Flash tank – liquid destination and CMV staging** Drawn: Flash vapour → CMV stages 1-2-3 in series; 70 °C liquid outlet left open. Question: Flash-tank liquid destination and pressure; confirm series staging.
- **H7** **Air-preheating coil – 85 °C limit on a 90 °C header** Drawn: One coil (TGE example) fed directly from the 90/70 °C header; no tempering arrangement. Question: How is the 85 °C limit best achieved, and how many coils are needed?
- **H8** **Thermal degasser – 4 bar or 1 bar steam?** Drawn: Steam inlet left unconnected. Question: Select 4 bar or 1 bar for the degasser.
- **H9** **Make-up water heater (12→80 °C) – heating medium** Drawn: Hot side left open. Question: Heating medium of the make-up heater.
- **H10** **LPF tank heating loops – pump and steam supply** Drawn: Plate exchangers on the tank loops; retained steam exchangers on standby with open steam/condensate nozzles.
- **H11** **Ultimate mode – well-water feed to the fermenters** Drawn: Only the post-fermenter discharge to DARSE is drawn.
- **H12** **Hot-water networks – local steam back-up equipment** Drawn: Not drawn (not described).
- **H13** **RCUA network battery limits** Drawn: Placeholder battery-limit flags for the 105 °C supply and 85 °C return.
- **H14** **Valve technology of the generic valves** Drawn: Generic valve symbol, DEXPI CustomOperatedValve 'technology to be selected'. Question: Select the valve technology for each valve function.

DATA TO BE COMPLETED ON DATASHEETS (NOT HOLDS)

- Hot-water process lines
- Internal loop PAC 2A/2B
- Gas boiler
- Equipment tags are given only for P-030..035, ECH-030..032, TK-001/002, K3/MESSO/GEA and the LPF tanks
- No nozzle numbers or nozzle sizes are given for any equipment (except DN50 on the TGE coil)
- No design pressures, relief devices/sizing, safety thresholds (other than 7.5 bar eff.), valve failure positions, material classes or instrument ranges are giv...
- PAC 3A/3B are future equipment with no capacity, regime or connection data
- Failure position (fail open / close / retain) of the control valves is not specified and was not assumed
- Measuring ranges and alarm/trip settings of the generated instruments are not specified and were not assumed
- Nozzles added for drain, vent and level tapping have no nozzle number and no size

REV	DATE	DESCRIPTION
A	2026-10-23	Draft generated from the specification – issued for engineering review

PROJECT
Projet Décarbonation LPF & BSST

TITLE
Décarbonation énergétique LPF & BSST – draft P&ID

CONTENT
Piping & instrumentation diagram – complete scope on one sheet

DRAWING No.	REV.	SHEET
PID-LPF-BSST-001	**A**	**1 of 1**
STATUS	SCALE	DATE
DRAFT – FOR REVIEW	**NTS**	**2026-10-23**

STANDARDS: ISO 10628-2 symbols · ISA 5.1 tags · DEXPI 1.3 / Proteus XML
SOURCE: Cahier des Charges P&ID Consolidé.pdf

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